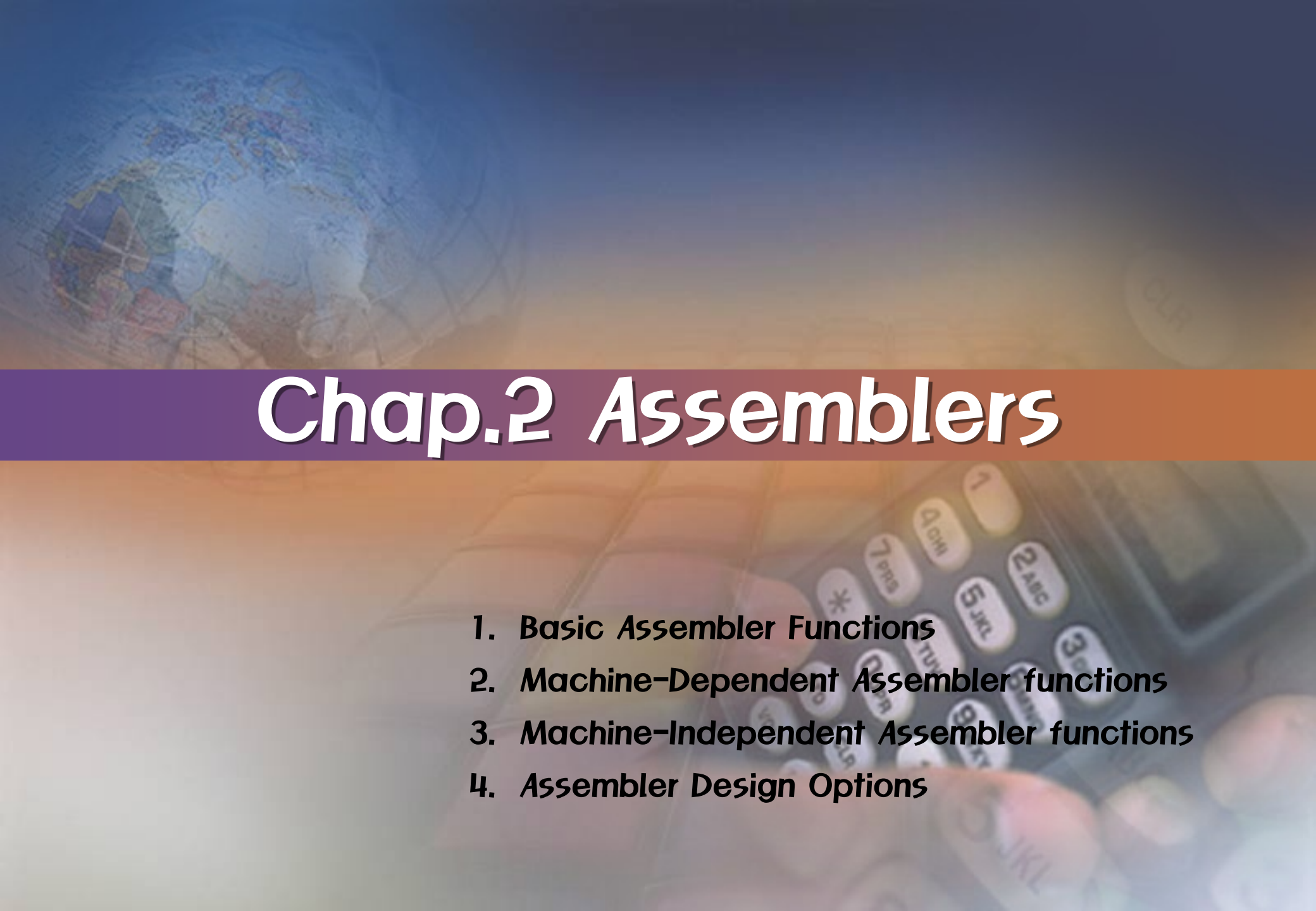


The background of the slide features a composite image. On the left, there is a stylized globe with a grid pattern, showing various colors like blue, green, and red. On the right, there is a close-up of a calculator's keypad, with buttons for numbers 1 through 9, 0, and various mathematical symbols like *, /, %, and =. The overall background has a warm, orange-brown gradient.

Chap.2 Assemblers

1. **Basic Assembler Functions**
2. **Machine-Dependent Assembler functions**
3. **Machine-Independent Assembler functions**
4. **Assembler Design Options**

수업 목표

- *Assembler*의 기본 개념 및 기능에 대해 이해한다.
- 예제를 통해 기본 개념과 기능 학습을 확인한다.

Assemblers

- Discuss the design and implementation of Assemblers
- Any assembler must perform
 - Translation of assembly codes to machine language
 - Assignment of machine addresses to symbolic labels

Basic Assembler Functions

- SIC assembler language program (Fig. 2.1)
- Assembler Directives
 - **START**: specifies name and starting address for the program
 - **END**: indicates the end of the source program and (optionally) specifies the first executable instruction in the program
 - **BYTE**: generates character a hexadecimal constant, occupying as many as bytes as needed to represent the constant
 - **WORD**: generates one-word integer constant
 - **RESB**: reserves the indicated number of bytes for a data area
 - **RESW**: reserves the indicated number of words for a data area

Example of a SIC program (Fig 2.1)

5	COPY	START	1000	COPY FILE FROM INPUT TO OUTPUT
10	FIRST	STL	RETADR	SAVE RETURN ADDRESS
15	CLOOP	JSUB	RDREC	READ INPUT RECORD
20		LDA	LENGTH	TEST FOR EOF (LENGTH = 0)
25		COMP	ZERO	
30		JEQ	ENDFIL	EXIT IF EOF FOUND
35		JSUB	WRREC	WRITE OUTPUT RECORD
40		J	CLOOP	LOOP
45	ENDFIL	LDA	EOF	INSERT END OF FILE MARKER
50		STA	BUFFER	
55		LDA	THREE	SET LENGTH = 3
60		STA	LENGTH	
65		JSUB	WRREC	WRITE EOF
70		LDL	RETADR	GET RETURN ADDRESS
75		RSUB		RETURN TO CALLER
80	EOF	BYTE	C'EOF'	
85	THREE	WORD	3	행 번호는 참조를 위해서 사용하고, 프로그램의 일부는 아니다.
90	ZERO	WORD	0	
95	RETADR	RESW	1	
100	LENGTH	RESW	1	LENGTH OF RECORD
105	BUFFER	RESB	4096	4096-BYTE BUFFER AREA

Example of a SIC program (Fig 2.1)

.으로 시작하는 문장은 항상 주석문이다.

110	.			
115	.	SUBROUTINE TO READ RECORD INTO BUFFER		
120	.			
125	RDREC	LDX	ZERO	CLEAR LOOP COUNTER
130		LDA	ZERO	CLEAR A TO ZERO
135	RLOOP	TD	INPUT	TEST INPUT DEVICE
140		JEQ	RLOOP	LOOP UNTIL READY
145		RD	INPUT	READ CHARACTER INTO REGISTER A
150		COMP	ZERO	TEST FOR END OF RECORD (X'00')
155		JEQ	EXIT	EXIT LOOP IF EOR
160		STCH	BUFFER,X	STORE CHARACTER IN BUFFER
165		TIX	MAXLEN	LOOP UNLESS MAX LENGTH
170		JLT	RLOOP	HAS BEEN REACHED
175	EXIT	STX	LENGTH	SAVE RECORD LENGTH
180		RSUB		RETURN TO CALLER
185	INPUT	BYTE	X'F1'	CODE FOR INPUT DEVICE
190	MAXLEN	WORD	4096	

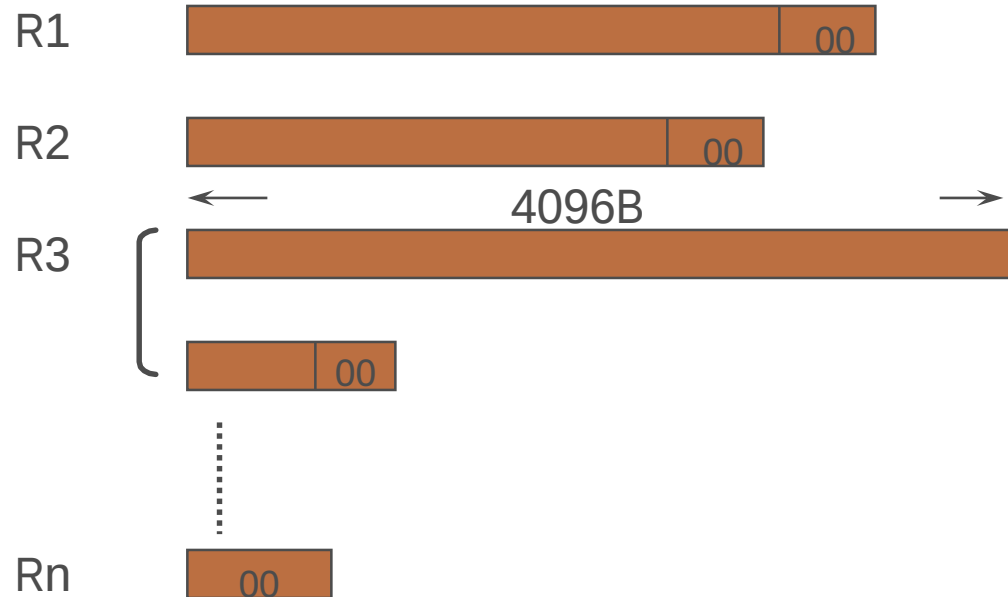
Example of a SIC program (Fig 2.1)

피연산자 뒤에, x를 추가하는 것은 인덱스 주소지정 방식을 나타낸다.

195	.			
200	.	SUBROUTINE TO WRITE RECORD FROM BUFFER		
205	.			
210	WRREC	LDX	ZERO	CLEAR LOOP COUNTER
215	WLOOP	TD	OUTPUT	TEST OUTPUT DEVICE
220		JEQ	WLOOP	LOOP UNTIL READY
225		LDCH	BUFFER,X	GET CHARACTER FROM BUFFER
230		WD	OUTPUT	WRITE CHARACTER
235		TIX	LENGTH	LOOP UNTIL ALL CHARACTERS
240		JLT	WLOOP	HAVE BEEN WRITTEN
245		RSUB		RETURN TO CALLER
250	OUTPUT	BYTE	X'05'	CODE FOR OUTPUT DEVICE
255		END	FIRST	

Program Logic (1/2)

Record



: zero-length record


output

final output

E	O	F
---	---	---

{ input device code : F1_H
output device code : 05_H

Program logic (2/2)

- **Data transfer (RD, WD)**

- A buffer is used to store record
- Buffering is necessary for different I/O rates
- The end of each record is marked with a null character (00_{16})
- Buffer length is 4096 Bytes
- The end of the file is indicated by a zero-length record

- **Subroutines (JSUB, RSUB)**

- RDREC, WRREC
- Save link (L) register first before nested jump

A Simple SIC Assembler

- (1) Convert mnemonic codes to their machine language equivalents
 - E.g. translate **STL** to **14** (line 10)
- (2) Convert symbolic operands to their equivalent machine addresses
 - E.g. translate **RETADR** to **1033** (line 10)
- (3) Build the machine instructions in the proper format
- (4) Convert the data constants specified in the source program into their internal machine representations
 - E.g. translate **EOF** to **454F46** (line 80)
- (5) Write the object program and the assembly listing

이 동영상 교과목 자료는
GJ-RIP 사업의
지자체-대학 협력기반 지역혁신 사업비로 개발되었음.



Chap.2 Assemblers